

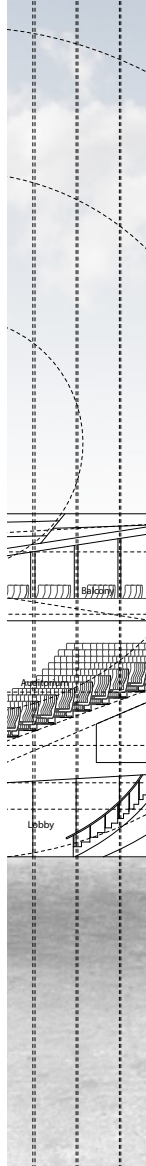
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Green House



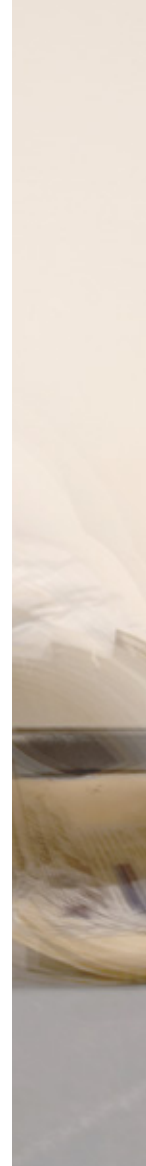
the  
Playground



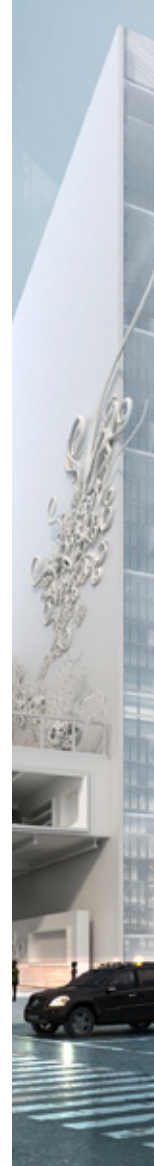
Theater  
with Two  
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4101  
Laclede



Snail  
Rocker



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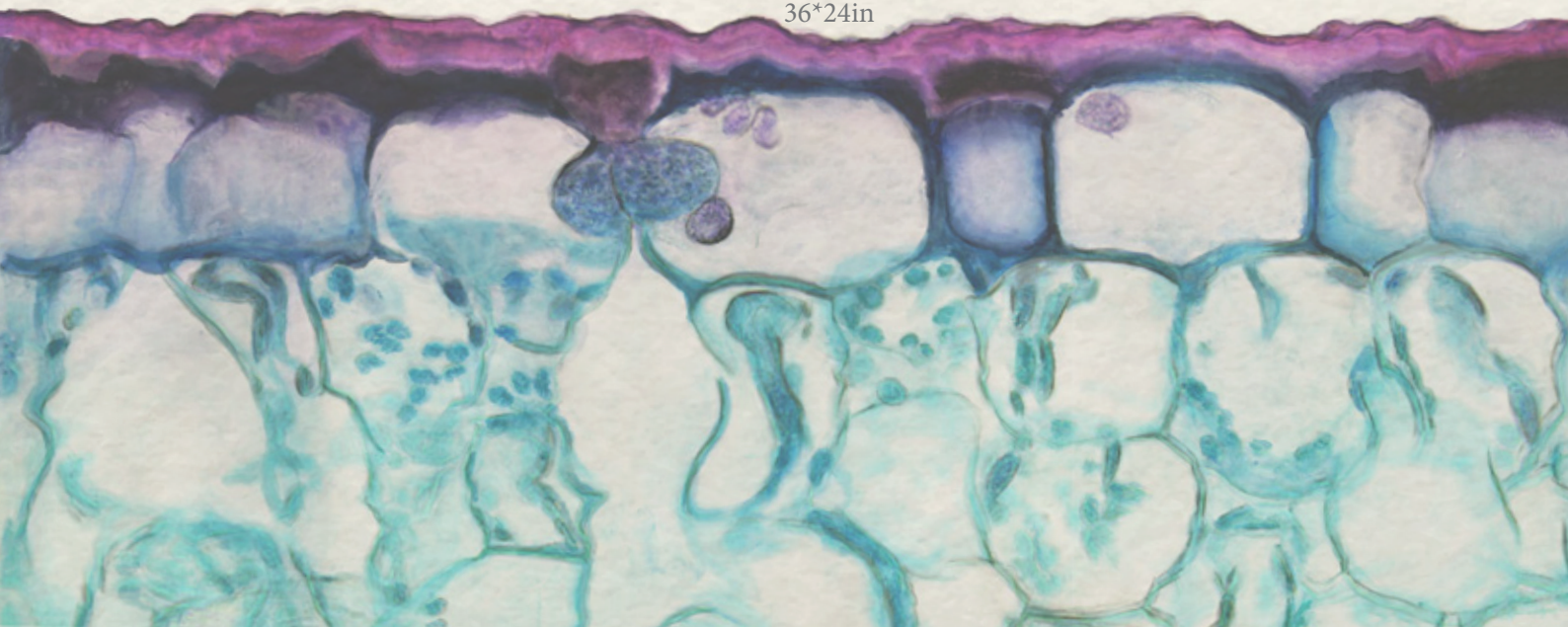


Echeveria elegans  
(Mexican Snowball)



Adult Echeveria and  
Propagation Stages,  
Graphit and Watercolor.

Echeveria Cells  
under Microscope  
Watercolor  
36\*24in



Greenhouse | Bathhouse

Studio 212 | 2018 Spring  
Jesse Vogler

This project focuses on the thermal and programmatic adjacencies within the building, as well as the urban and social relationships with a specific site. Starting from analytical researches on selected plants, I designed the Eco-Unit based on the botanical features and environment, and then design the bathhouse and greenhouse with a deeper investigation in Eco-Unit.

Eco Unit



The opening on the top  
allows plant to grow freely  
without any obstruction

The structural skin system not  
only provides support to the  
plant but also leave growing  
space for the baby plants

Pot with cone-shaped facilities

Before designing the actual Greenhouse, We started out with Eco Unit for single plant. According to the survival requirements of my plant, I customized a Eco Unit for her, which could sustain itself without further attention.

My Echeveria survived for the whole semester and even grew taller than it was planted in the Eco Unit.



Eco Unit



Pot and Echeveria



The cone-shaped skin structure  
plugs in onto the pot without any  
other connections necessary.  
Cone-shaped skin

The skin could be easily put on and taken off from the flower pot. Since the skin itself is strong enough to provide protection to the plant inside, I would therefore focus on this feature of the structure.



Verb List | sunlight and water

glow

glitter

beam

hit

contact

dash

spray

plash

splatter

groove

pipe

transmit

drip

trickle

leak

drop

**Illuminate**

**Shine**

**Gleam**

**Touch**

**Reflect**

**Splash**

**Penetrate**

**Channel**

**Drain**

**Dribble**

**Flood**

lighten

elucidate

brighten

enlighten

glare

glint

echo

mirror

reproduce

permeate

puncture

drill

drip

sap

deplete

exhaust

deluge

rush

overflow

7

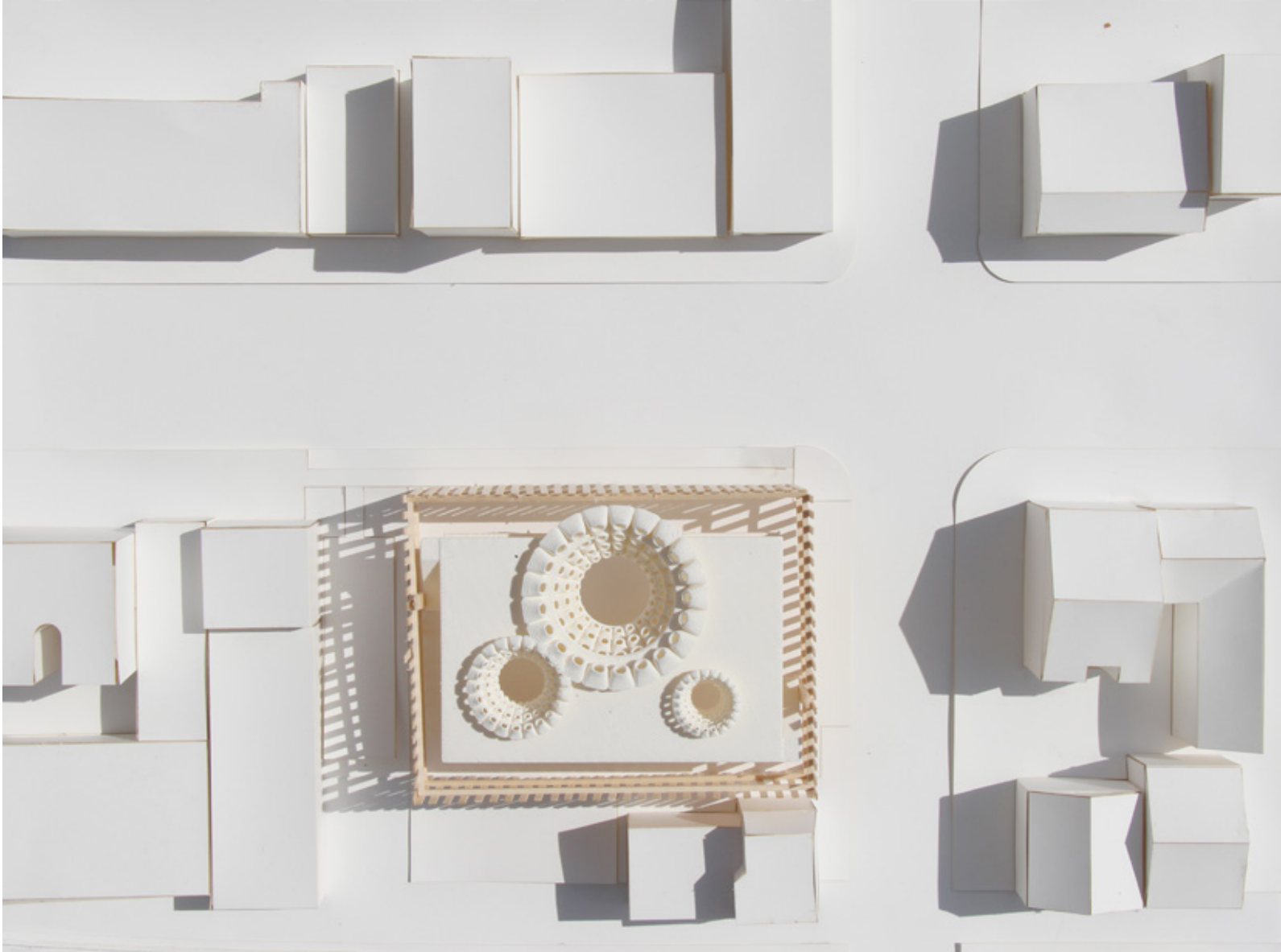
Model 6  
Flattened Collage



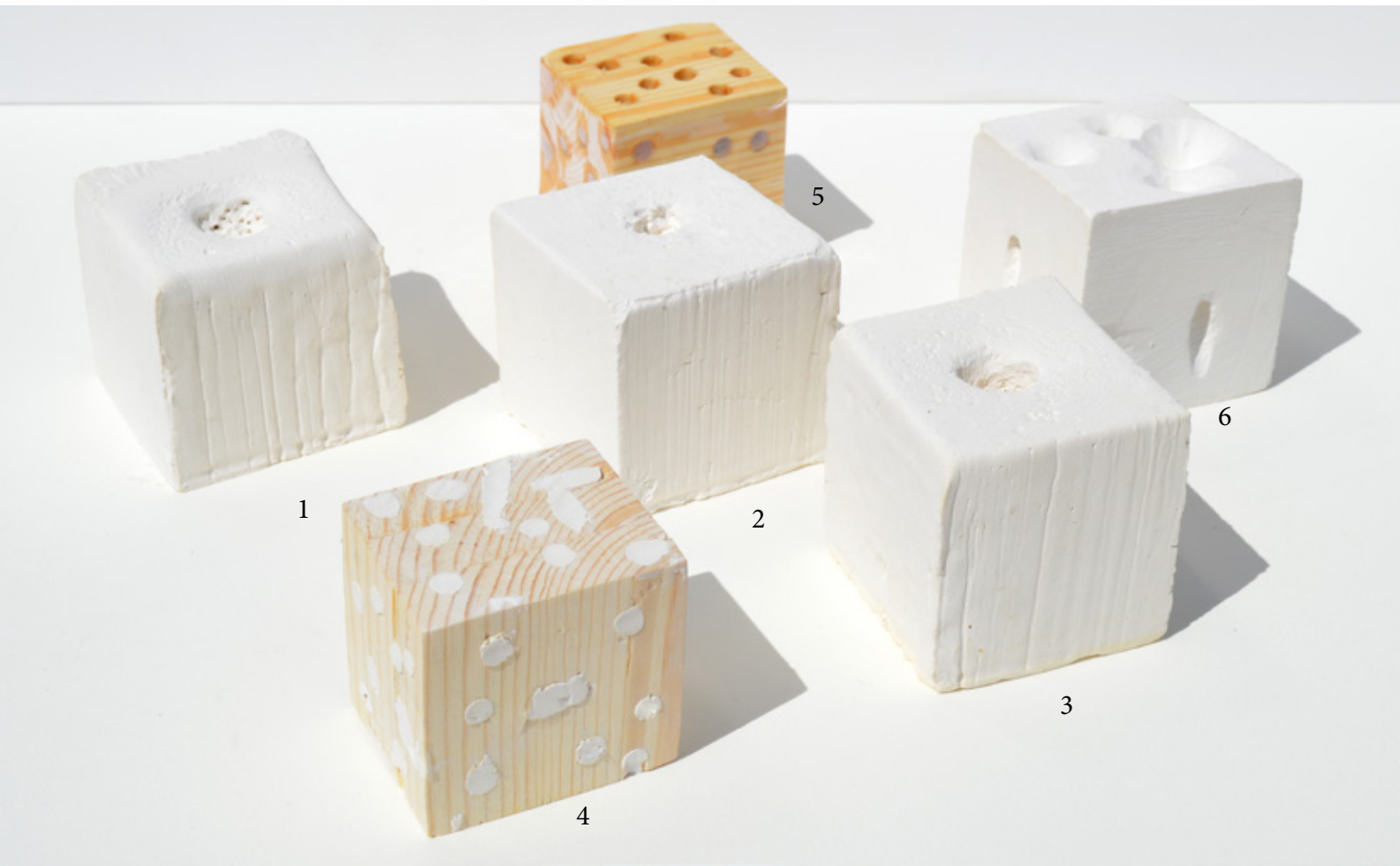
The common verb that both sunlight and water could used to describe is “to penetrate”.

So I did a series of concept models regarding to this verb using different materials and techniques.

Instead of building the block out of skeletal struc-  
ture, I rather take out some material and let the  
negative space be the primary structure.



1'=1/16" Scale model with  
Site



1 | 2 | 3

Three identical plaster cubs placed under dripping water for dif-  
ferent length to test out how water could shape other materials.

4 | 5

Wood cubes with hole at different angles, displaying the surface  
condition of the negative space.

6 |

Using conclusions drawn for the previous 5 models, this model  
shows how different holes at different angles could perform in a  
building.

7 |

Negative space created by the same geometry used in the Eco-  
Unit



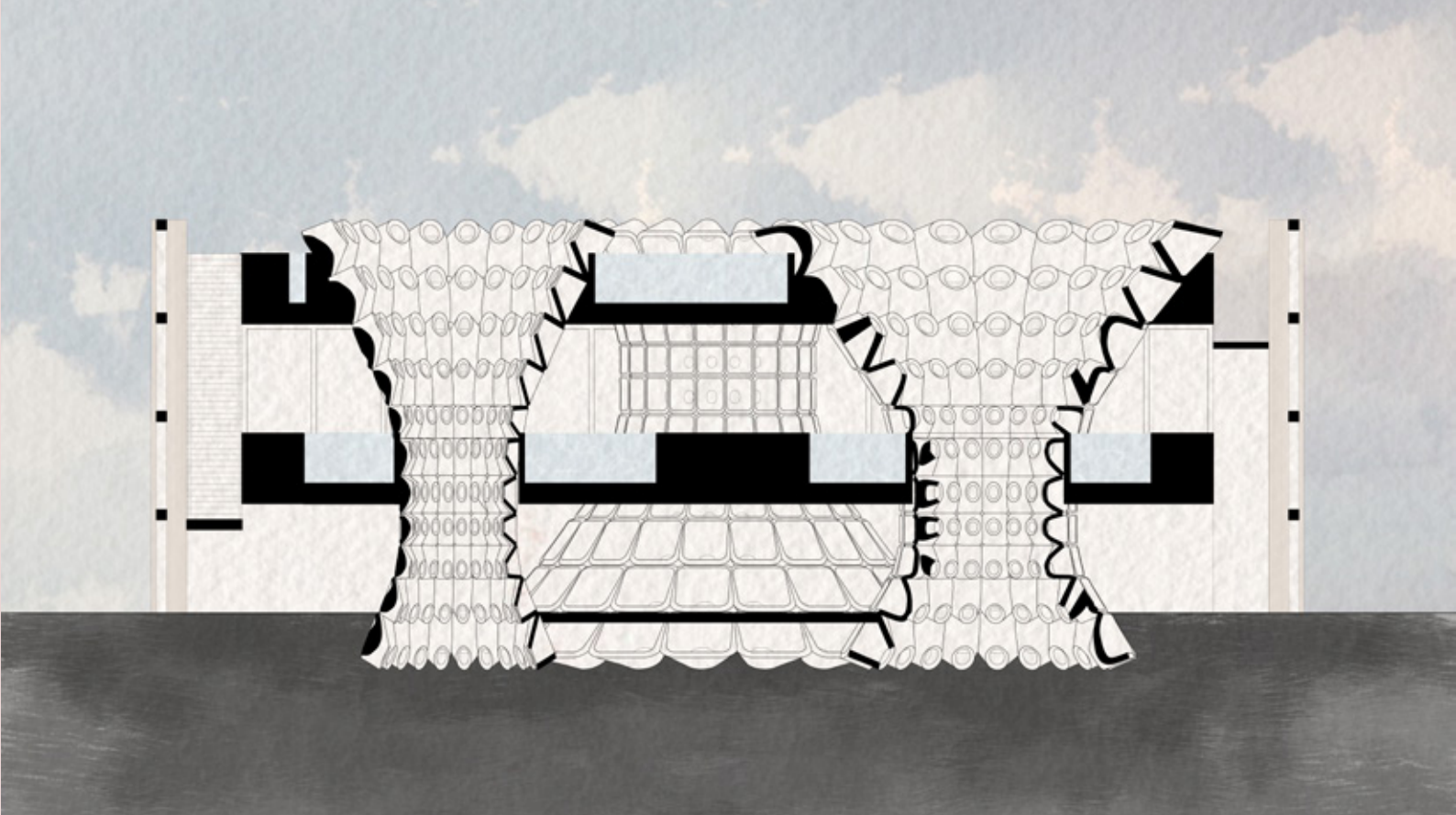
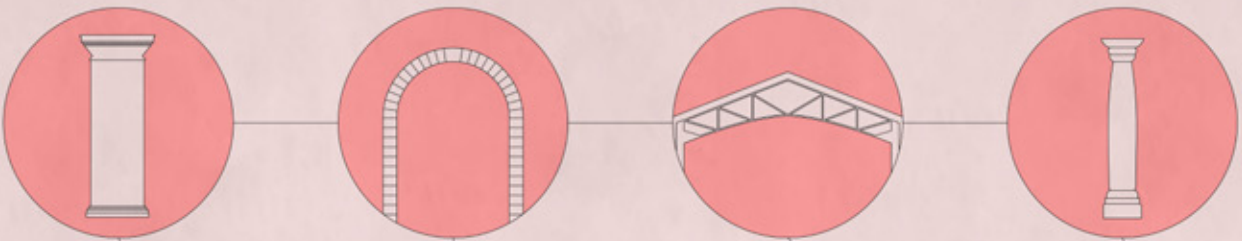


# Site

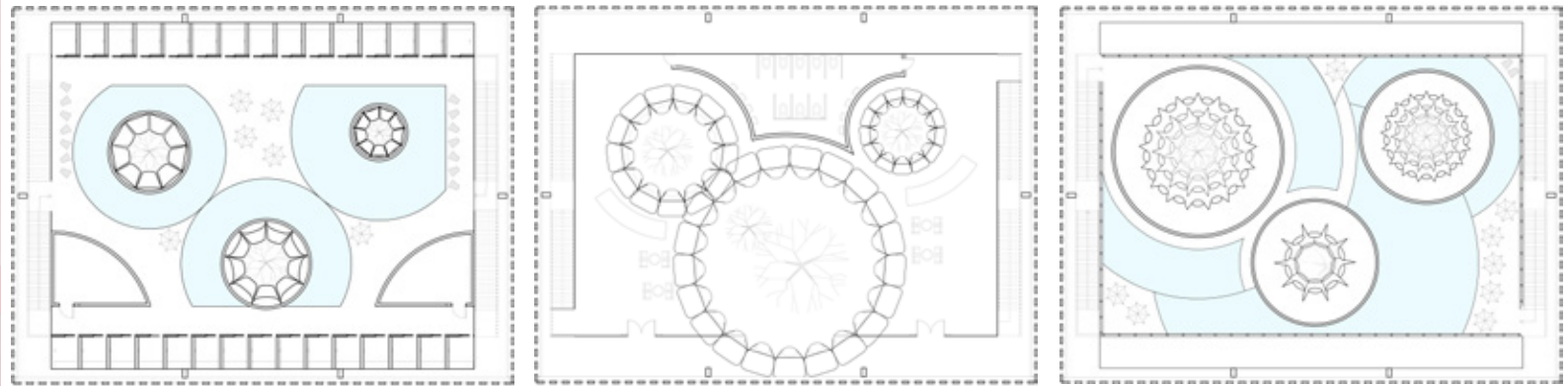
The site is in the historic Soulard neighborhood and is a corner, urban lot. In the process of investigating the site, I found that there are several different types of support system used by different forms of buildings. In the process of figuring out the evolution in support systems of the Domino house and the Sendai Mediatheque, my project primarily focuses on the support solely.



Support Systems on Site  
1/32"=1'0"



Section Drawing



Floor plans



Interior Render





5,710 children who crossed the Mediterranean last year were unaccompanied. A total of 25,846 children made the crossing.



-Victor, 5

Victor was rescued from the Mediterranean Sea after being separated from his mother. Victor was detained in Surman detention center for nearly two months.



-Kamir, 9

"They arrested us and brought us into the Zawia prison. No food. No water. They beat us every day. No doctor, no medicine."



TRIPOLI

BERLIN

TOBRUK

WARSAW

PRAGUE

SABHA

MUNICH

VIENNA

AL QATRUN

BUDAPEST

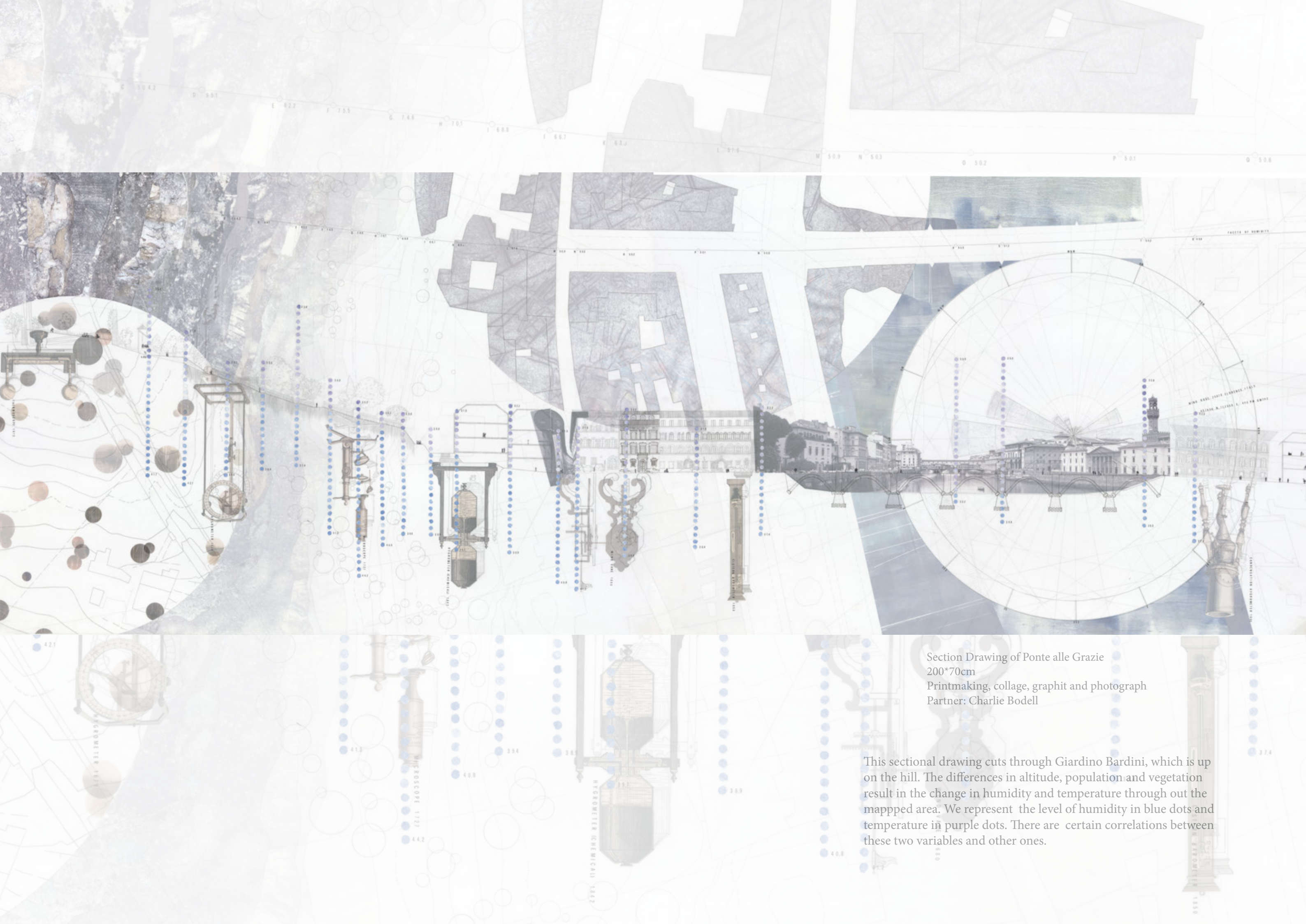
KOSICE

KUFRA

SZEGED

An estimated 4,579 people died crossing the Mediterranean between Libya and Europe last year alone, of which over 700 were children.





Section Drawing of Ponte alle Grazie  
200\*70cm  
Printmaking, collage, graphit and photograph  
Partner: Charlie Bodell

This sectional drawing cuts through Giardino Bardini, which is up on the hill. The differences in altitude, population and vegetation result in the change in humidity and temperature through out the mapped area. We represent the level of humidity in blue dots and temperature in purple dots. There are certain correlations between these two variables and other ones.





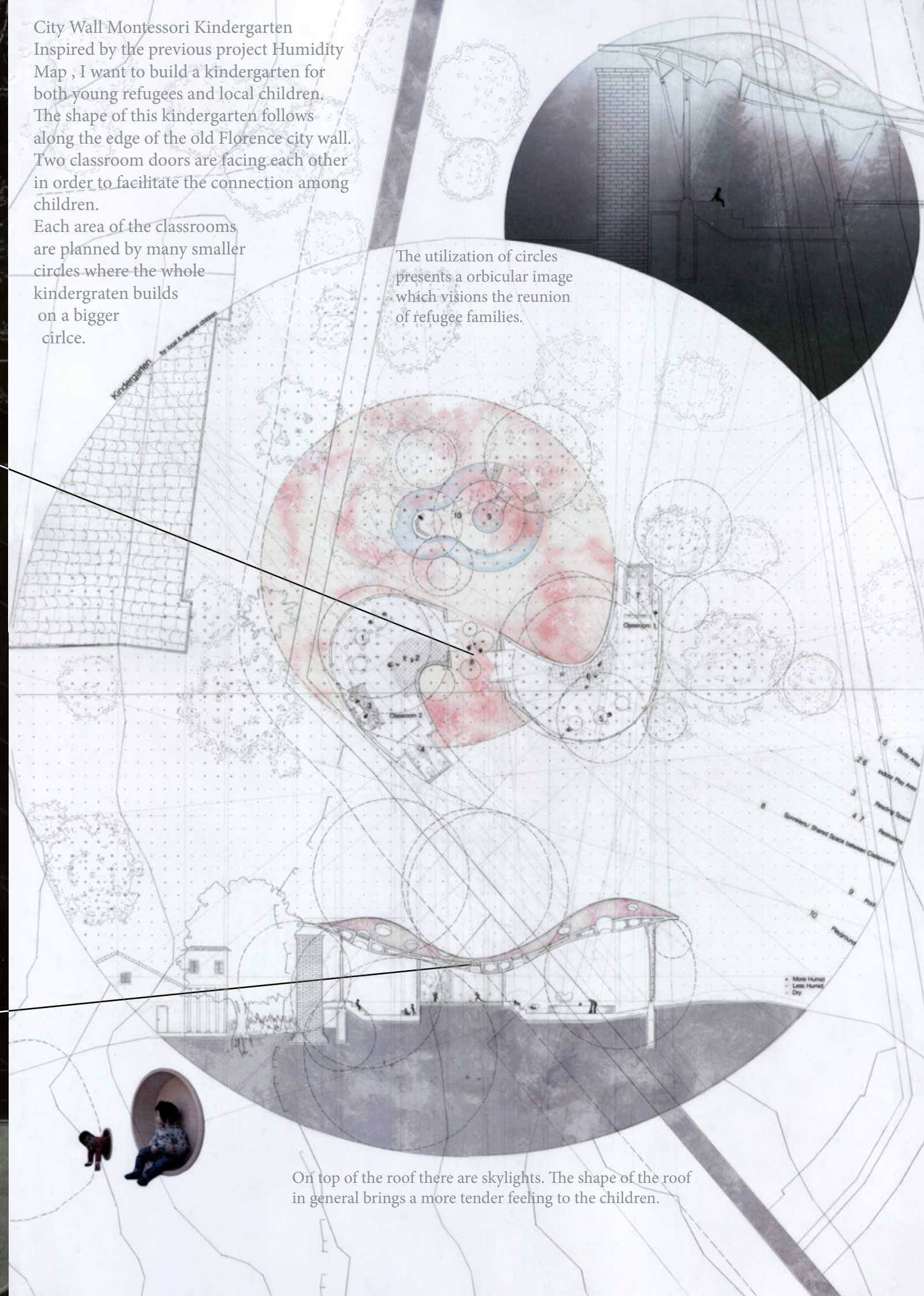
Plan

Section

City Wall Montessori Kindergarten  
 Inspired by the previous project Humidity Map , I want to build a kindergarten for both young refugees and local children. The shape of this kindergarten follows along the edge of the old Florence city wall. Two classroom doors are facing each other in order to facilitate the connection among children.

Each area of the classrooms are planned by many smaller circles where the whole kindergraten builds on a bigger circlce.

The utilization of circles presents a orbicular image which visions the reunion of refugee families.



On top of the roof there are skylights. The shape of the roof in general brings a more tender feeling to the children.



# Observatory

Studio III | 2016 Fall  
Igor Marjanovic

The design from an negative space drawing of the organic pattern, and then it went to the basswood and wire model, and after that, the observatory for the nature.

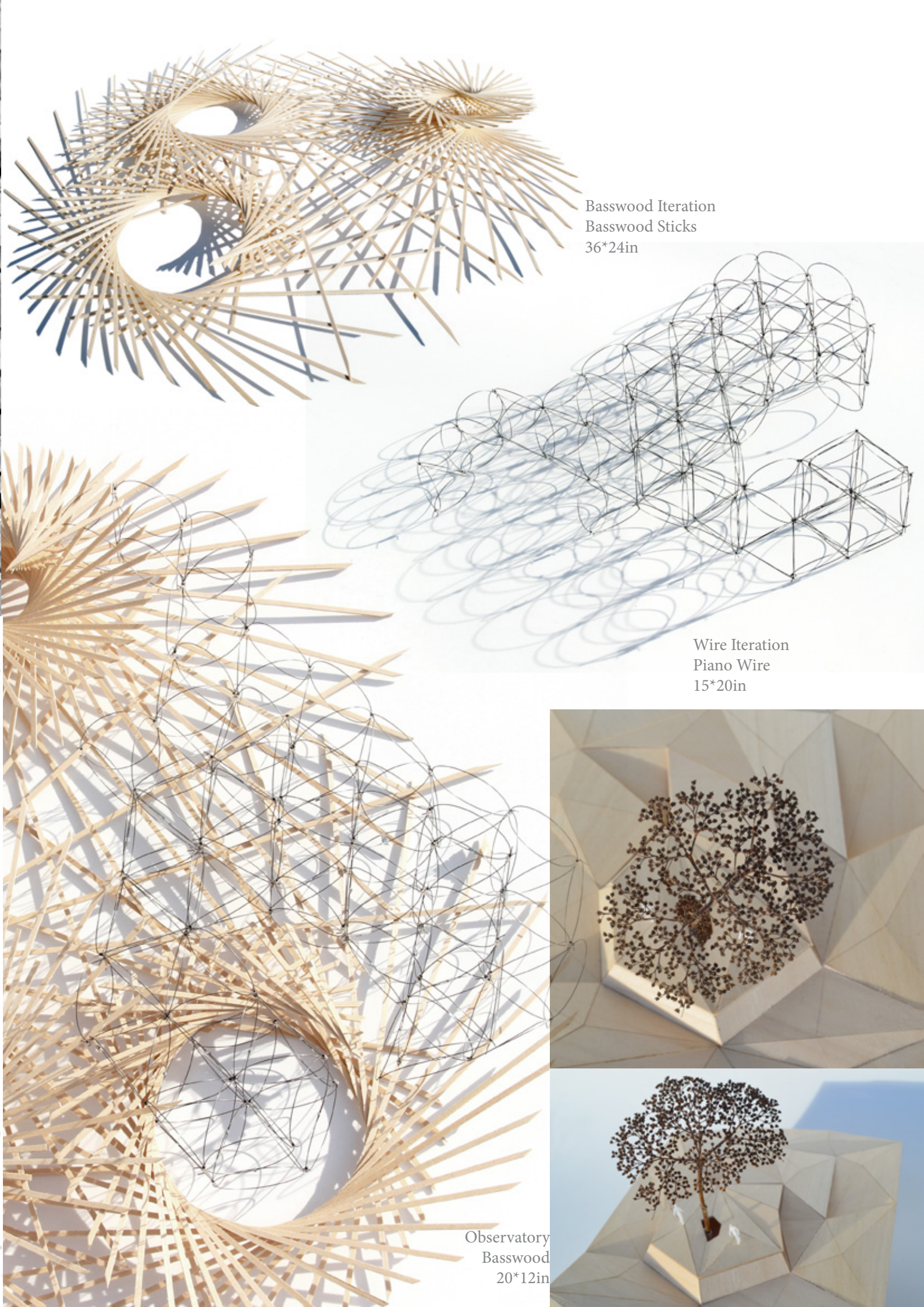
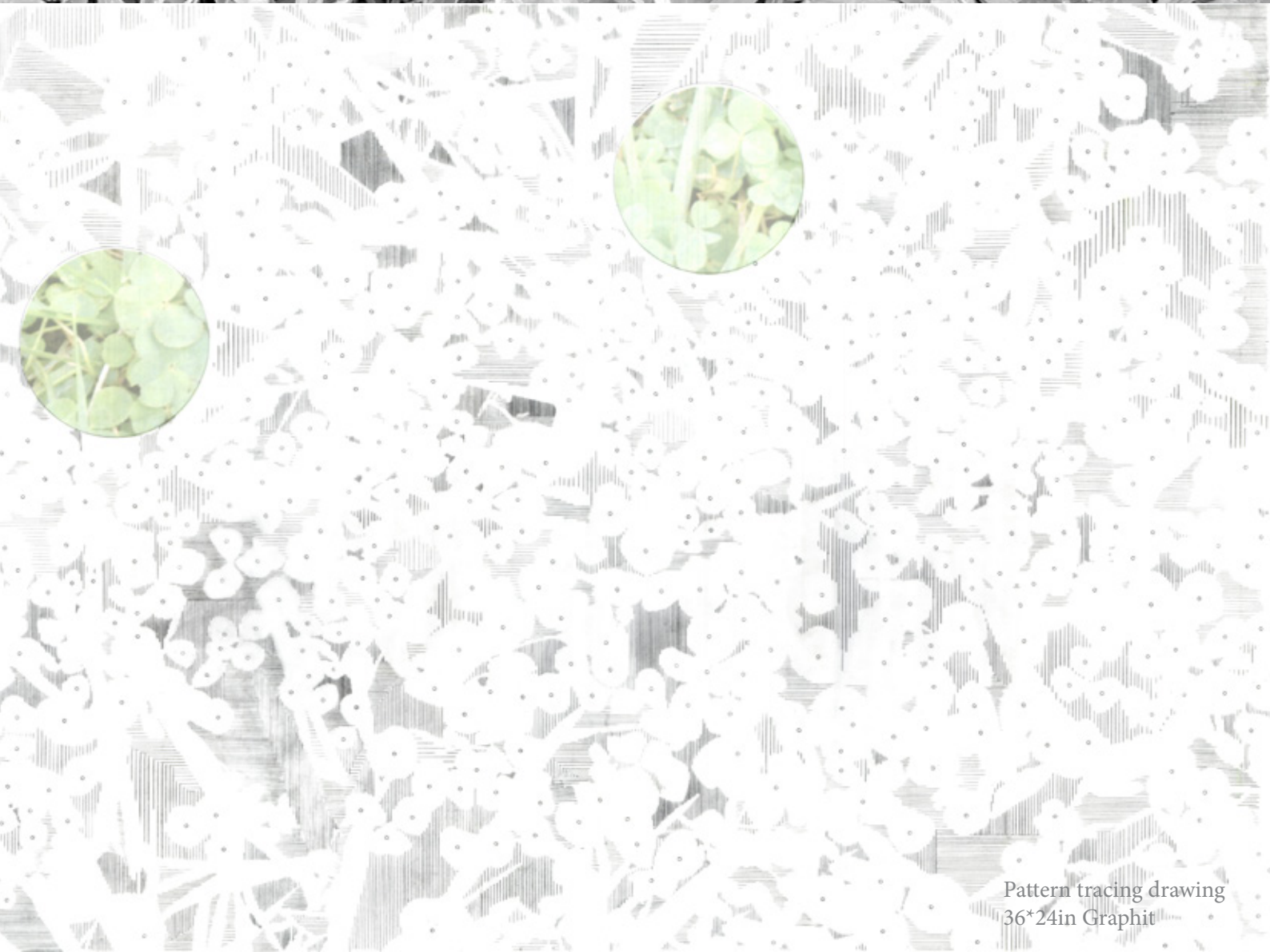
Natural Pattern  
Photograph  
Site: St. Louis Forest Park

Pattern tracing drawing  
36\*24in Graphit

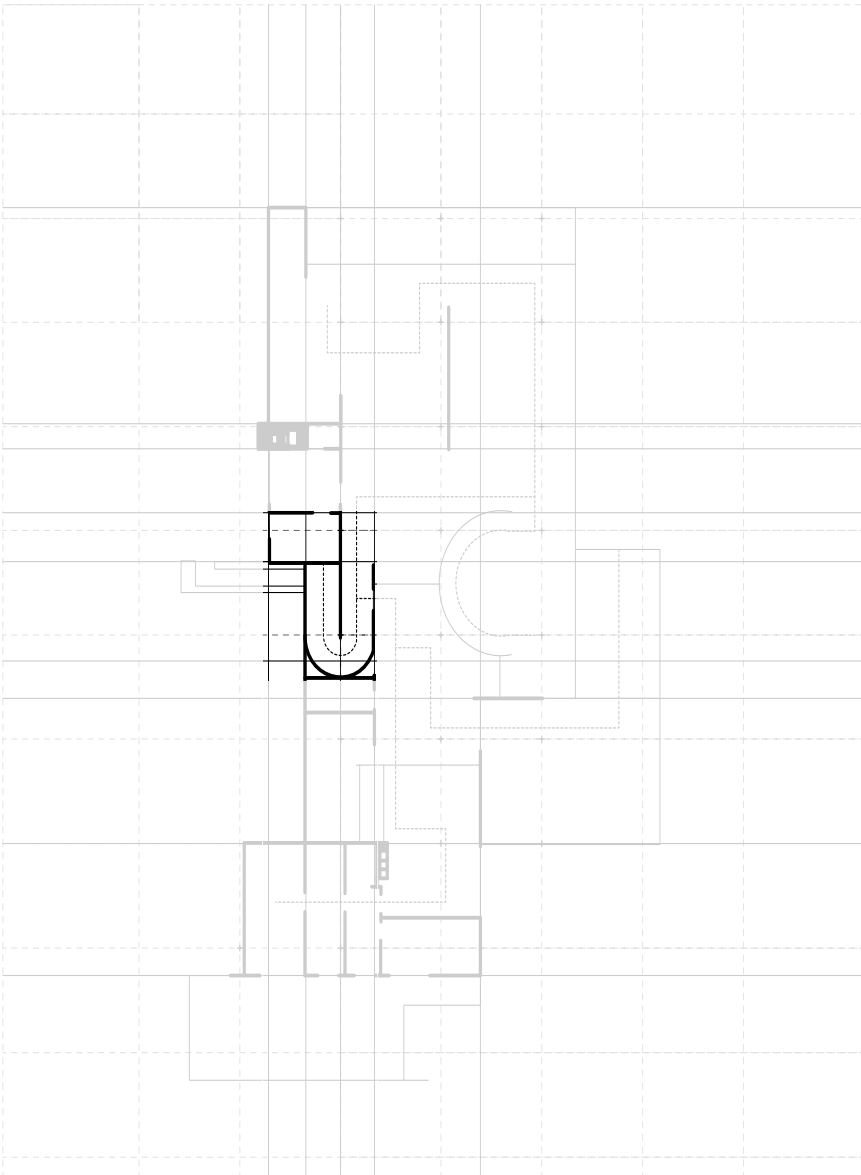
Basswood Iteration  
Basswood Sticks  
36\*24in

Wire Iteration  
Piano Wire  
15\*20in

Observatory  
Basswood  
20\*12in







Tugendhat House  
Mies Van der Rohe  
Brno, Czech Republic

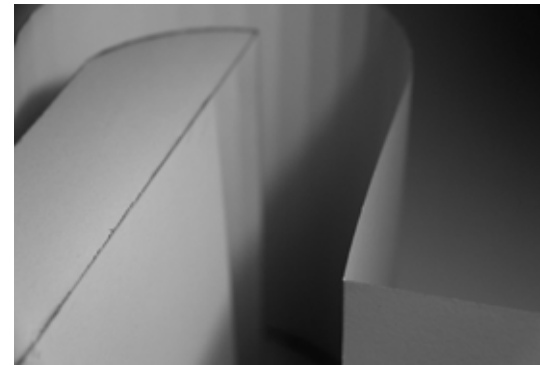
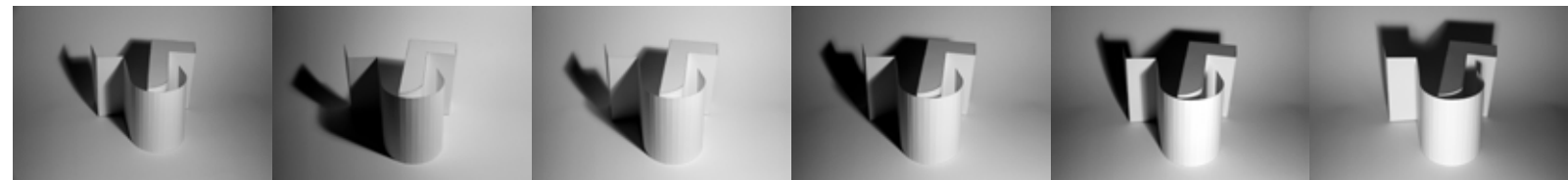
## Theater with Two Characters

Studio 211 | 2017 Fall  
Nathaniel Elberfeld

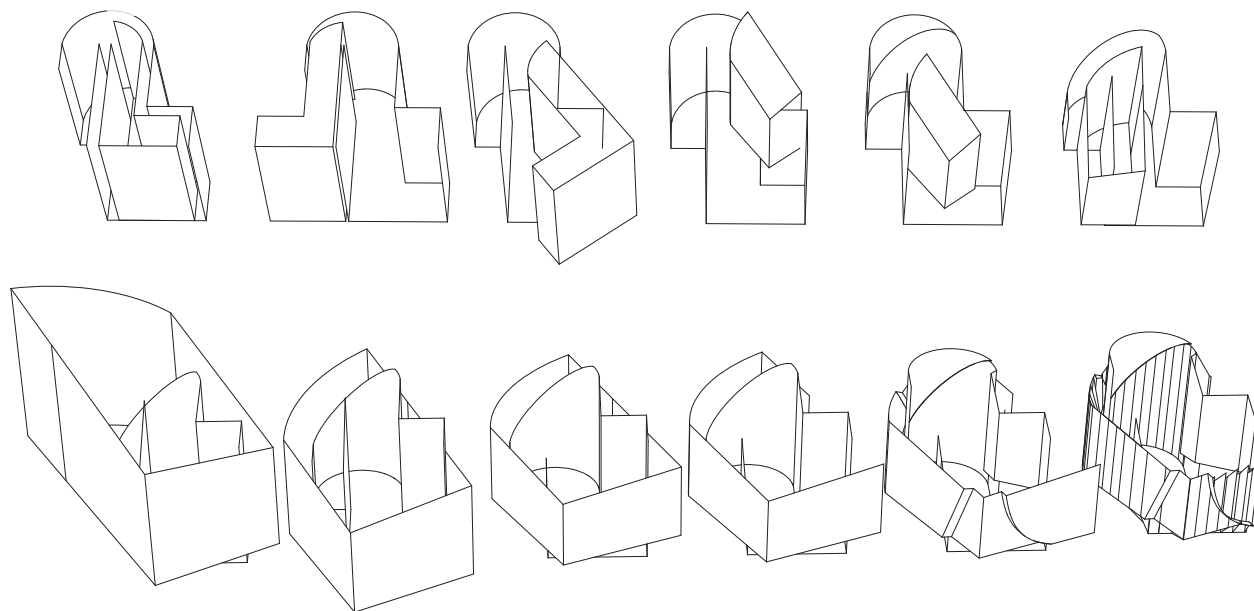
The project of theater is inspired by the “scars” on the walls, creating a space by pushing into the porche which is provided as the primary circulation.

Starting from a study prototype from Mies Van der Rohe, the study model derives from the original plan to two separate parts, and these parts progress to a more reasonable model.

The light study is able to grant the theater two characters: one with shadow and the other one with light.



1/16”=1’-0” scale model, White Bristol  
Initial shadow study provides a good reference for the visible “scars” portained by light. Two characters, light and shadow, intertwine with each other to create a distinctive boundary.

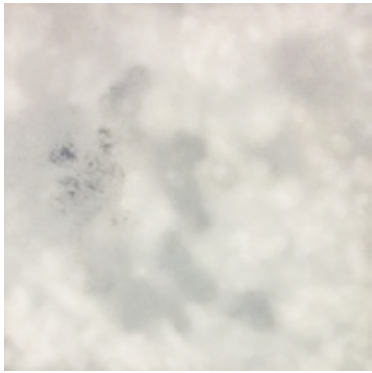


Form Transformation





Barcelona Pavilion,  
Barcelona, Spain.

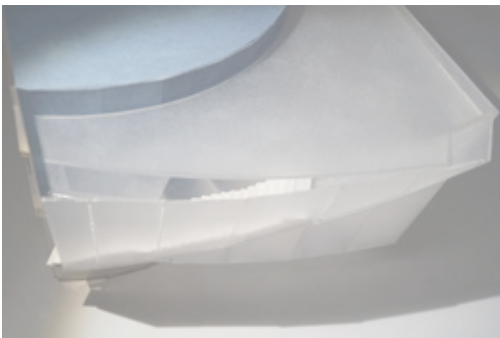
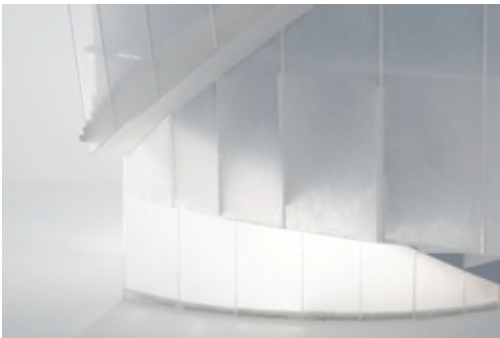
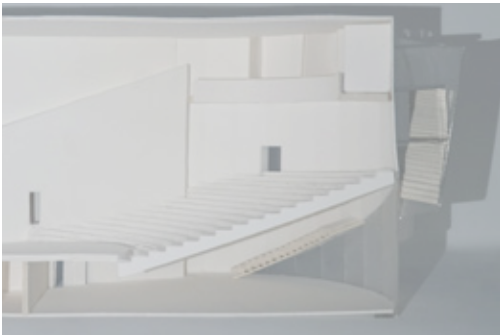
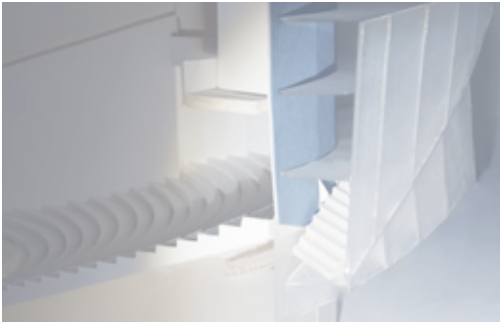


Derived Pattern Scan  
I

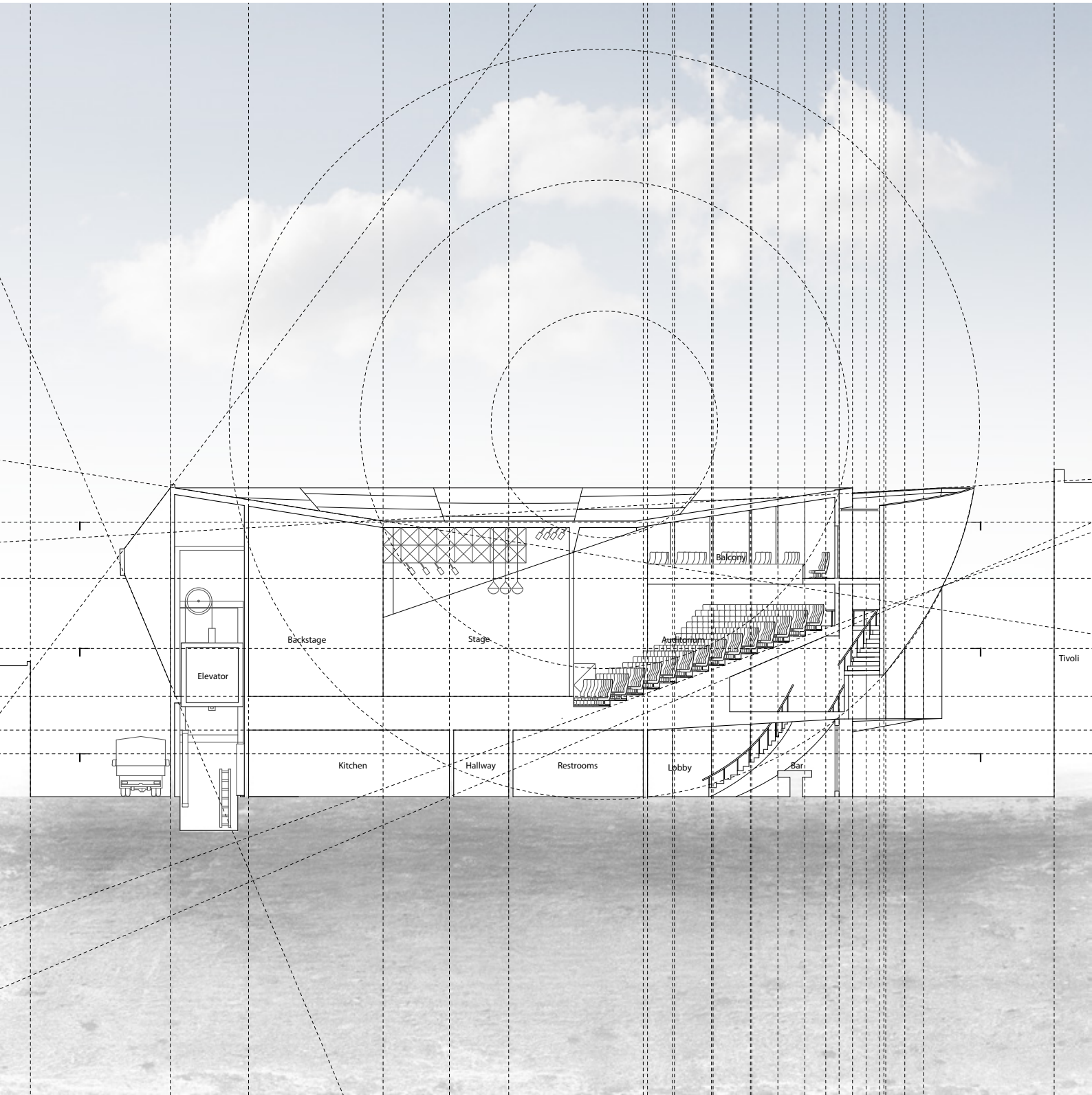


Derived Pattern Scan  
II

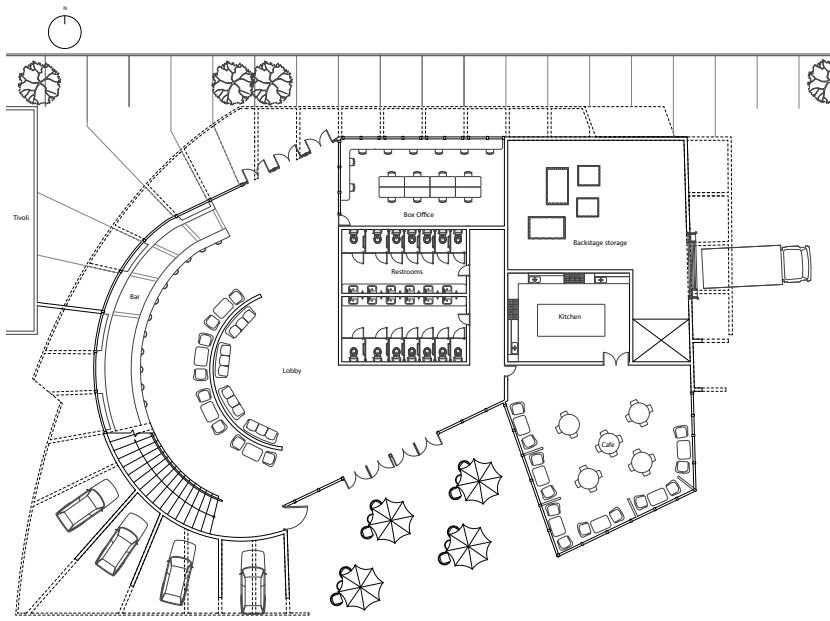
The theater is designed according to the arrangement of the sidewalk and goes along with its orientation. The circulations all happen in the porche in between the outer shell and the inner space. This hierarchy distinguishes private and public also provides natural sun-light during the day for the theater. At night, people could see the light in the theater from the street. The two light conditions characterizes this amphitheater.



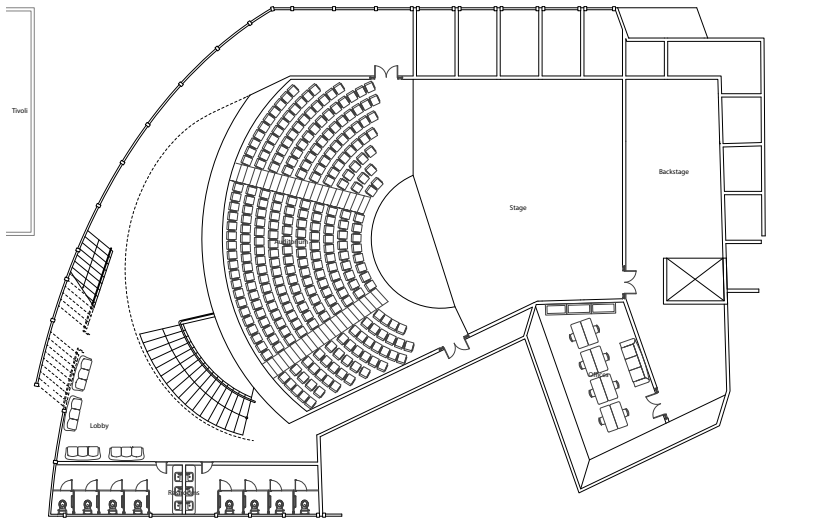
Theater Section Model Photos  
1/8" = 1'-0" Scale



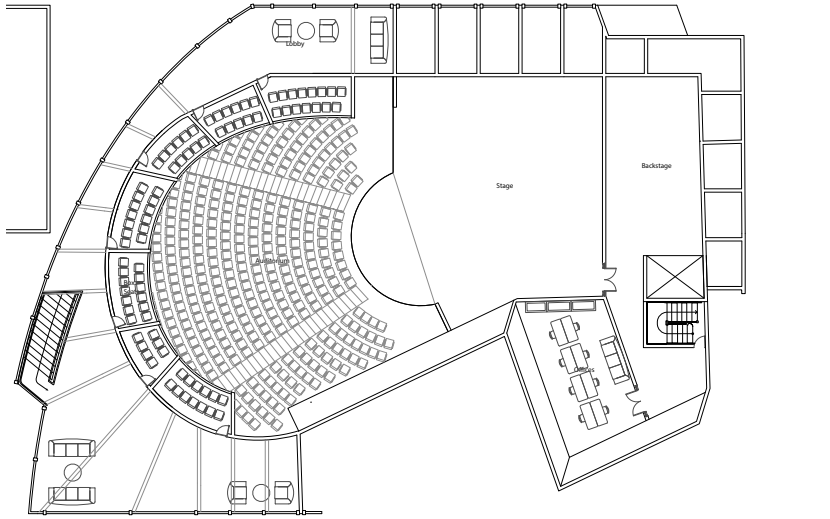
Section Drawing



Ground Floor



Second Floor



Third Floor

Floor Plans



4101 Laclede  
Building Systems | 2018 Fall  
Ross Welch  
Partners: Zinan Chi  
Wentao Guo

Located at one of the busiest area in St. Louis, this modern, luxury residence are surrounded by restaurants, bars, pharmacies and post office. In addition to convinient transportation, the Central West End neighborhood is right next to Cortex, BJC Hospital, Euclid Shopping Corridor, Forest Park, Washington University Medical School, SLU, and many other Public Services.

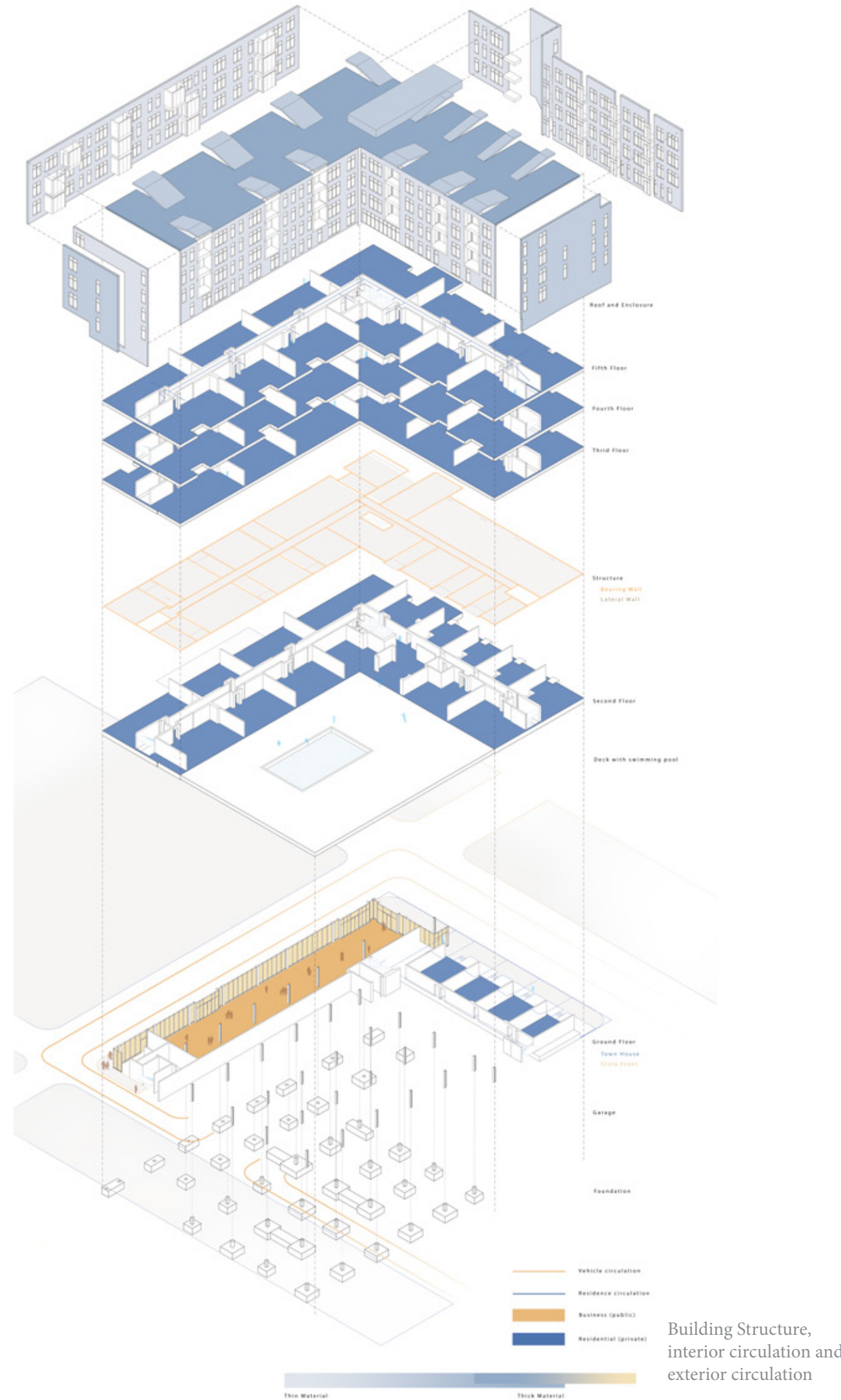
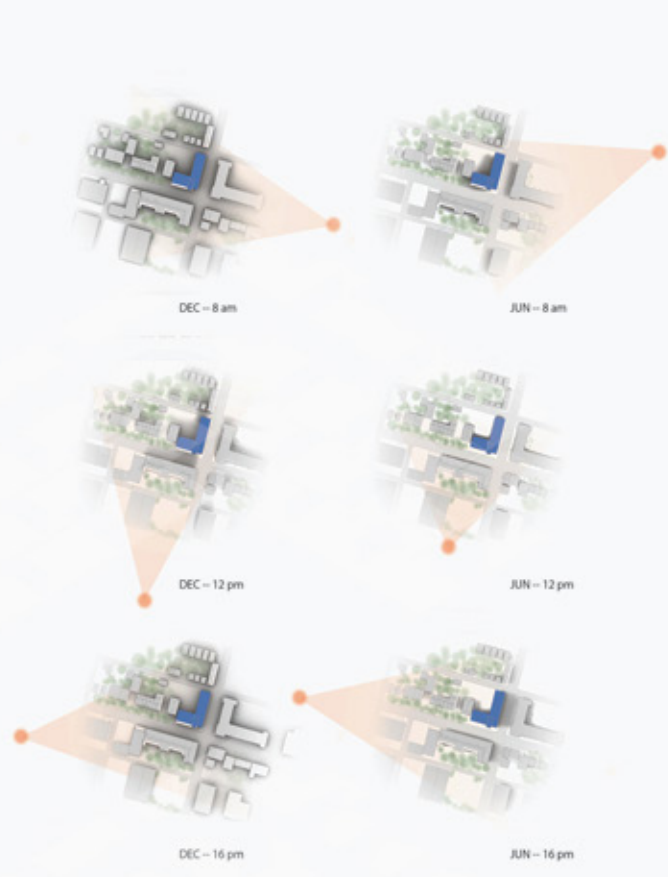
Surroundings



Wind Study

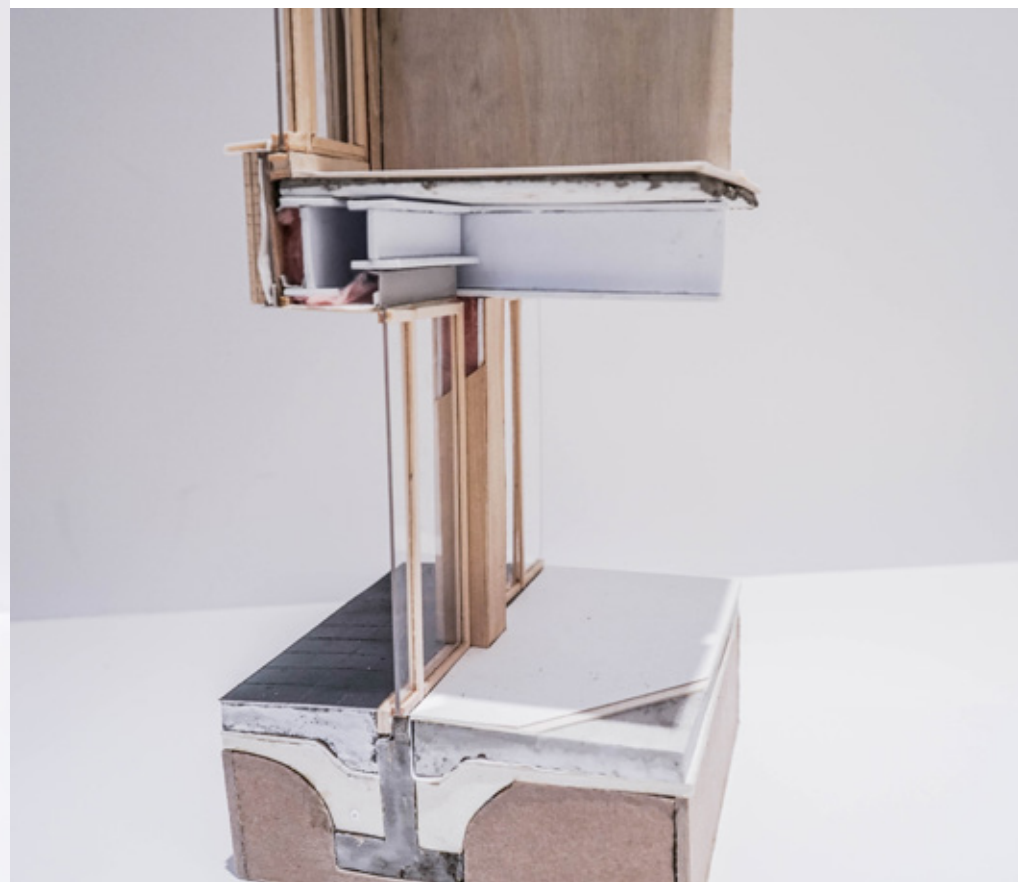
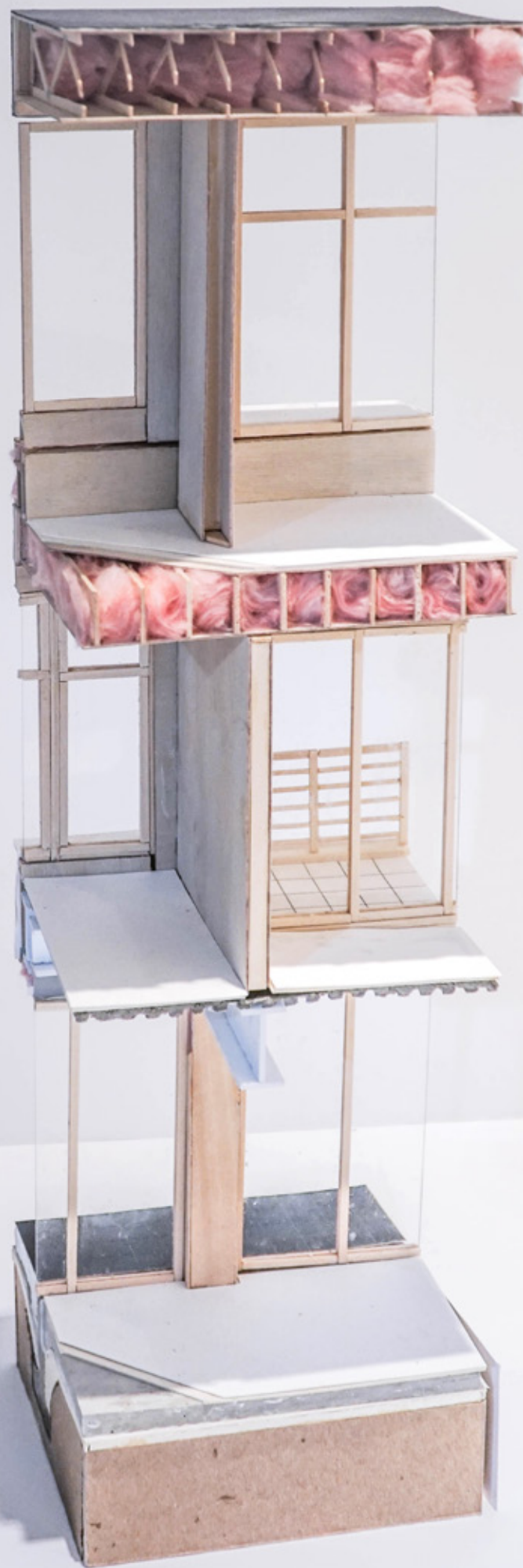


Shadow Study



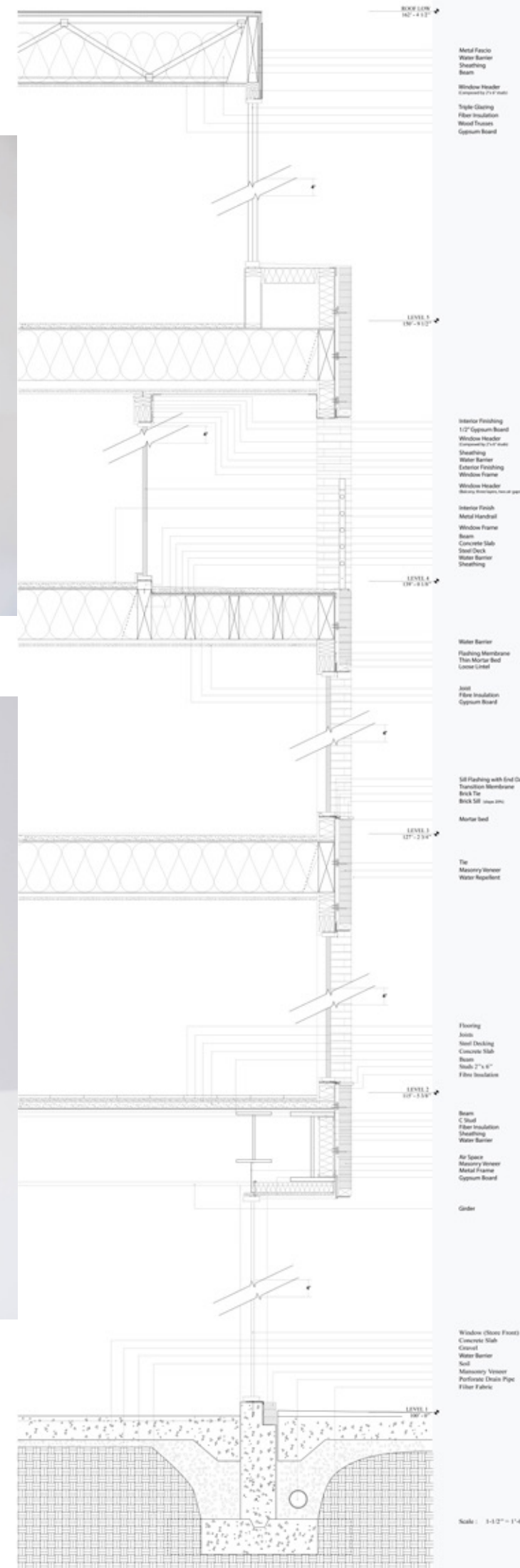
Building Structure, interior circulation and exterior circulation





Model details

Wall section scale model  
3/2"=1'-0"



### Section Study

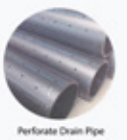
4101 Laclede is built in a combination of wood frame and steel frame. This choice of structural frame is mainly under consideration of building function. 4101 Laclede is constructed in steel frame for retailers on the first floor taking the advantages of the long span of steel frame. From the second floor to the fifth, the residential part is constructed in wood frame.



Plywood



Stud 2"x6"



Perforate Drain Pipe



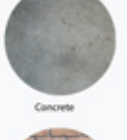
Vapor Retarder



Cypsum



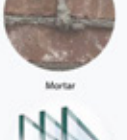
Fiber Insulation



Concrete



Masonry Veneer



Mortar



Triple Glazing



Gravel



Steel Decking

Scale: 3/4"=1'-0"

Wall section drawing



# Snail Rocker

Furniture Design | 2017 Fall  
Lindsey Stouffer

Starting from the snail shell, the rocker has been modified according to its function and shape. Bend wood technique creates many more possibilities to the realm of furniture design and architecture.



Structural Model



Final Model

